Name: ____________________________ Date: ________________ Period: _________

Primary Objective

I will be able to...

1. **LO 3.7.A** — Construct a sinusoidal model from a data set by computing amplitude, midline, period, and phase shift, and validate the model against known data values. **Construct sinusoidal models using amplitude, midline, period, and phase shift.**
2. **LO 3.7.A** — Use a sinusoidal model to predict output values and identify the contextual domain over which the model is meaningful. **Use sinusoidal models to predict values over the contextual domain.**

Vocabulary & Key Concepts

sinusoidal model: A function of the form $f(x) = A\sin(B(x - C)) + D$ (or cosine form) used to represent a **periodic** real-world quantity.

amplitude: $A = \dfrac{\max - \min}{2}$; measures how far the function deviates from its **midline**.

vertical shift / midline: $D = \dfrac{\max + \min}{2}$; the horizontal line $y = D$ about which the function **oscillates**.

period: The **smallest** input-value interval between consecutive maxima (or minima) in the data. Equals $\dfrac{2\pi}{B}$.

phase shift: C; shifts the graph **right** when $C > 0$. Determined by comparing a data maximum to the standard model maximum location.

contextual domain: The set of input values over which the model is **meaningful / valid** given the real-world context.

Hook

Entry Question: The table below shows average monthly high temperatures (°F) for a city over one year.

Month x	1	2	3	4	5	6	7	8	9	10	11	12
Temp $T(x)$ (°F)	38	42	54	65	74	82	86	84	75	63	49	40

Q1. Which month has the maximum temperature? **7 (July)** Which has the minimum? **1 (January)**

Q2. What kind of function could model this data? **A sinusoidal (sine or cosine) function, because the data rises, falls, and would repeat periodically over successive years.**

LO 3.7.A — Building a Sinusoidal Model: Four Steps

We use the model $T(x) = A\sin(B(x - C)) + D$ where:

- $A =$ amplitude, $D =$ midline (vertical shift),
- $B = \dfrac{2\pi}{k}$ ($k =$ period), $C =$ phase shift.

Step 1 — Find the Period k

The period equals the input-value distance between consecutive **maxima** (or consecutive **minima**).
In the temperature table, consecutive maxima occur at months **7** and **7** (next year), so $k \approx \mathbf{12}$.

$B = \dfrac{2\pi}{k} = \dfrac{2\pi}{12} = \dfrac{\pi}{6} \approx 0.524$

Step 2 — Compute Amplitude and Midline

$\max = \mathbf{86} \quad \min = \mathbf{38}$

$A = \dfrac{\max - \min}{2} = \dfrac{\mathbf{86} - \mathbf{38}}{2} = \mathbf{24}$

$D = \dfrac{\max + \min}{2} = \dfrac{\mathbf{86} + \mathbf{38}}{2} = \mathbf{62}$

The midline represents **the average of the hottest and coolest monthly temperatures (62°F)** in context.

Step 3 — Determine the Phase Shift C

For $T(x) = A\sin(B(x - C)) + D$, the maximum occurs when $B(x - C) = \dfrac{\pi}{2}$, so $x = C + \dfrac{\pi}{2B}$.
The data maximum occurs at $x = \mathbf{7}$. Set equal and solve for C:

$C + \dfrac{\pi}{2B} = \mathbf{7} \quad \Rightarrow \quad C \approx 7 - \dfrac{\pi}{2 \cdot \pi/6} = 7 - 3 = 4$

Tip: If you use a **cosine** model $T(x) = A\cos(B(x - C)) + D$, the maximum occurs when $B(x - C) = 0$, so $C = x_{\max} = 7$ **directly**.

Step 4 — Write and Validate the Model

$T(x) = 24\sin\left(\frac{\pi}{6}(x - 4)\right) + 62$

Validate: substitute a known data point (e.g., $x = 1$, $T = 38$):
$T(1) \approx 24\sin(-\frac{\pi}{2}) + 62 = 24(-1) + 62 = 38 \quad$ Residual $= 38 - 38 = 0$

Contextual Domain: This model is valid for $1 \le x \le 12$ **(one calendar year for this location)**.

Guided Practice

Guided Practice. The table below shows the depth of water (in feet) at a harbor entrance over one tidal cycle (12 hours).

Time t (hr)	0	2	4	6	8	10	12
Depth $d(t)$ (ft)	5	9.5	13	9.5	5	0.5	5

(a) Identify max = **13** at t = **4** and min = **0.5** at t = **10**.

(b) Compute: $A = \dfrac{13 - 0.5}{2} = 6.25 \qquad D = \dfrac{13 + 0.5}{2} = 6.75$

(c) Estimate the period k = **12** hr. Then $B = \dfrac{2\pi}{k} = \dfrac{\pi}{6} \approx 0.524$.

(d) Identify the phase shift C, using a cosine model with maximum at $t = 4$. C = **4**.

(e) Write the model: $d(t) = 6.25\cos\left(\frac{\pi}{6}(t - 4)\right) + 6.75$

(f) Use your model to predict the depth at $t = 3$. $d(3) = 6.25\cos(\frac{\pi}{6}(3 - 4)) + 6.75 = 6.25\cos(-\frac{\pi}{6}) + 6.75 \approx 6.25(0.866) + 6.75 \approx 12.16$ **ft**

(g) Is $t = 15$ hr inside the contextual domain for one tidal cycle? Explain. **No. The contextual domain is $0 \le t \le 12$ (one tidal cycle). $t = 15$ is outside that range, though the model could extend if the tidal pattern continues.**